Conditional time series forecasting with convolutional neural networks

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Paper Summary: WaveNet Architecture

uWN — Unconditional WaveNet

- Forecasts next-day return using **stock features only** — no external market signals
- Stacked **dilated causal convolutions** with doubling dilation rates (d = 1, 2, 4, 8 ...)
- Receptive field grows **exponentially** with depth: r = 2^(L-1) * k — captures long-term history without long sequences
- **ReLU activation** + residual connections — simpler and faster than gated WaveNet, avoids vanishing gradients
- Far **fewer trainable parameters** than LSTM — shared weights across the feature map enable faster training
- Trained with **MAE loss, Adam optimizer**, and L2 regularization to prevent overfitting on noisy financial data

In our project: uWN takes 48 stock-level features (momentum, volatility, volume, oscillators) as input to rank stocks cross-sectionally.

cWN — Conditional WaveNet

- Extends uWN: additionally conditions on e**xternal time series** such as SPY returns and VIX levels
- Condition applied via a **separate parallel dilated convolution** summed into the main input path via a parametrized 1x1 skip connection
- Skip connection weights can go to zero — model **automatically discards uninformative conditions**
- Exploits **cross-series correlations** to reduce noise — especially useful with limited training history
- Easily extended to **multiple conditions** — each gets its own parallel filter, all summed at the first layer

In our project: cWN receives SPY daily returns and VIX as market-wide conditioning signals alongside each stock's own features.

Paper Summary: Key Results

Forecasting Accuracy

Lorenz Chaotic System

- cWN X-coord RMSE: 0.00174 vs uWN 0.00577 — 3x improvement from conditioning
- cWN beats the augmented LSTM benchmark (avg RMSE 0.00675) from Hsu (2017)
- Conditioning also reduces forecast variance — smaller std dev in RMSE across seeds

S&P 500 & Exchange Rates

- **uWN achieves the lowest MASE (0.685)** — better magnitude accuracy than LSTM (0.873) and VAR
- cWN boosts directional hit rate in period A (2008–10) by exploiting VIX–index correlations
- On 5 FX pairs: cWN consistently outperforms VAR and LSTM on MASE across all periods, including during the 2008 crisis

Why It Works Better

vs. LSTM

- **Parallel training**: convolutional operations are fully parallelizable; LSTM is inherently sequential
- **Fewer parameters**: shared weights across the feature map reduce overfitting risk
- **No vanishing gradient**: residual connections preserve gradient flow across deep layers

General Strengths

- **Noise resistant:** each layer discards high-frequency noise — similar to wavelet decomposition
- **Short-history friendly:** conditioning compensates by exploiting correlated series — no long price history needed
- **Regime adaptive:** does not overfit when macro correlations weaken — conditions are discarded automatically

Summary of Our Work

Data

- S&P 500 (~500 stocks, 2010–2025), daily OHLCV
- 48 features: momentum, volatility, volume, technical indicators, SPY & VIX market signals
- Walk-forward splits: 2yr train / 6mo val / 3mo test, rolling every quarter — no lookahead

Model

- uWN: stock features only (unconditional)
- cWN: stock features + SPY & VIX as separate condition convolutions (conditional)

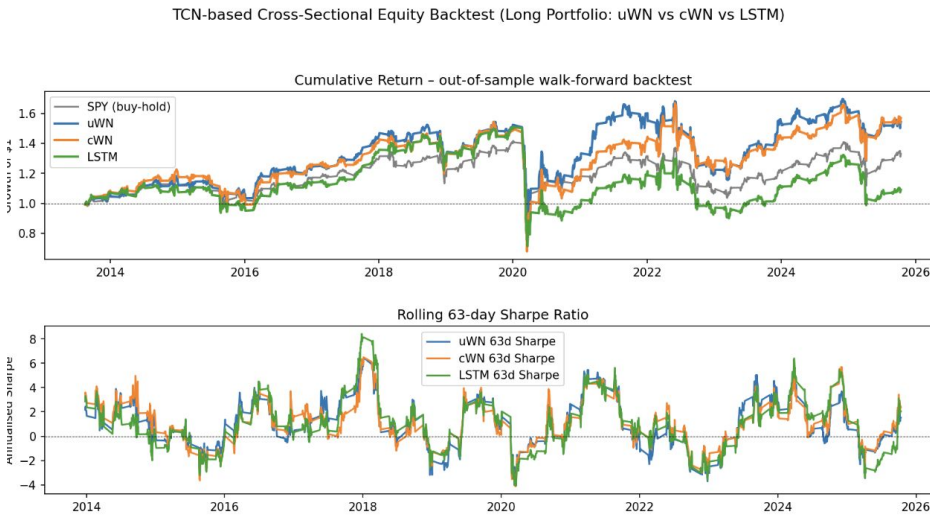
Strategy

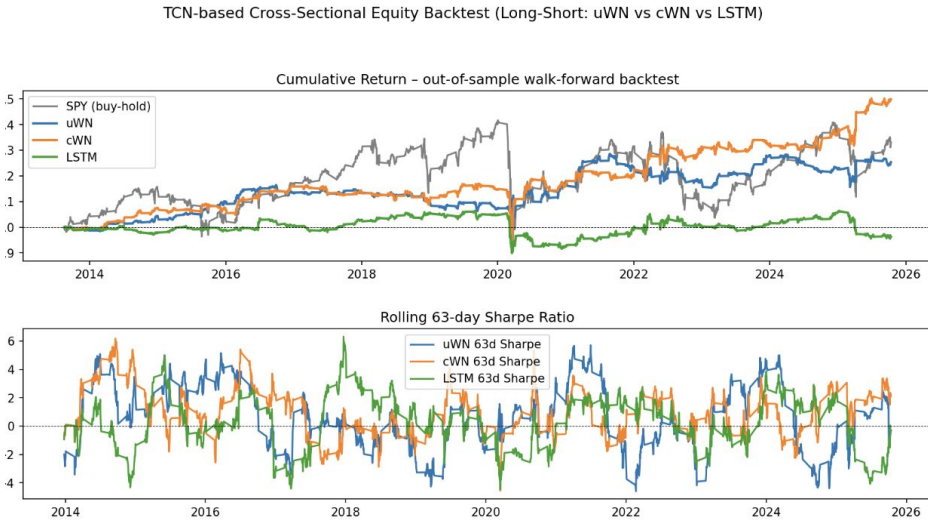
- Predict next-day return for every stock → rank cross-sectionally
- Long top 20%, short bottom 20% or long top 20%
- Rebalanced daily, 10 bps transaction cost per side applied

Backtesting

- Out-of-sample walk-forward backtest, 2013–2026
- Benchmarks: SPY buy-and-hold, LSTM
- Metrics: cumulative return, annualized Sharpe, max drawdown, hit rate

Backtesting Results & Comparison





- The **conditional WaveNet (cWN)** achieves the strongest results in the **long-short portfolio**
- The cWN delivers the **highest cumulative return** at the end of the backtest period
- The cWN **rolling Sharpe ratio is generally more stable**, indicating stronger risk-adjusted performance

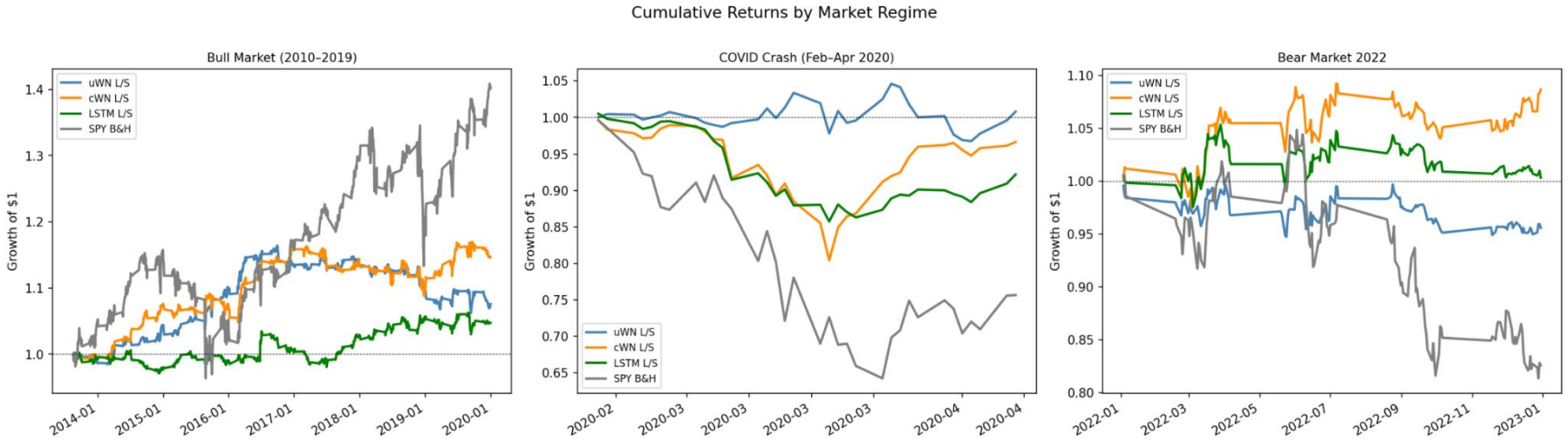
- Compared with uWN, cWN benefits from **conditioning on market-wide information**
- Compared with LSTM, cWN shows **better robustness in cross-sectional ranking**
- These results suggest that **conditional structure improves predictive power and portfolio construction quality**

Backtesting Results & Comparison

name	cum_return	ann_return	ann_vol	sharpe	max_dd	hit_rate	n_days
uWN (L/S)	25.4	3.5	7.3	0.476	-10.8	51.4	1666
cWN (L/S)	49.9	6.3	8.7	0.726	-21.0	51.3	1666
LSTM (L/S)	-3.7	-0.6	6.7	-0.085	-15.4	50.6	1666
uWN (Long)	54.4	6.8	24.2	0.280	-43.9	54.8	1666
cWN (Long)	56.9	7.1	25.6	0.275	-56.1	54.5	1666
LSTM (Long)	9.1	1.3	24.1	0.055	-52.7	54.6	1666
SPY buy-hold	33.08	4.42	18.93	0.233	-36.17	53.66	1666

- cWN is the strongest model in both long-short and long-only settings
- In the **long-short portfolio**, cWN achieves:
  - **Highest cumulative return:** 49.9%
  - **Highest annual return:** 6.3%
  - **Best Sharpe ratio:** 0.726
- cWN also has the **highest hit rate** among the three long-short models
- In the **long-only portfolio**, cWN again ranks first with:
  - **Cumulative return:** 56.9%
  - **Annual return:** 7.1%
- cWN Long (56.9%) and uWN Long (54.4%) **outperform SPY buy-and-hold**
- Overall, the results indicate that **conditional market information improves predictive power and portfolio performance**

Regime Sensitivity

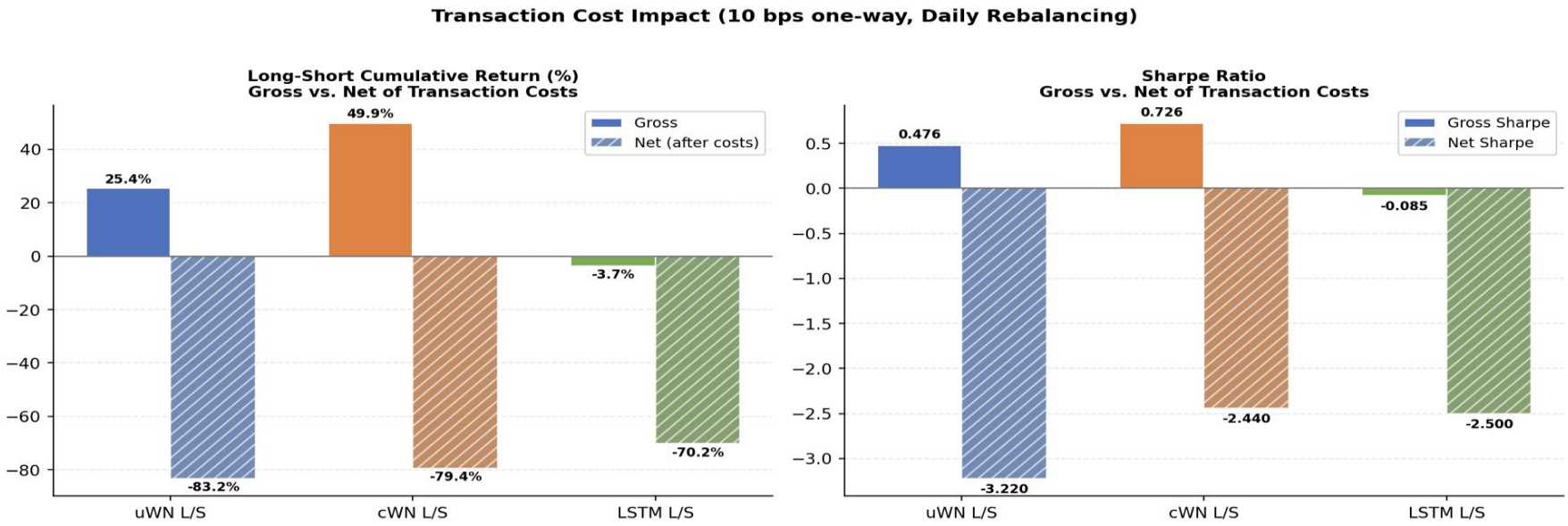


Sharpe Ratio Comparison

	uWN L/S	cWN L/S	LSTM L/S	SPY B&H
Bull	0.538	**0.849**	0.375	0.816
COVID	0.362	**-0.505**	-2.517	-2.276
Bear	-0.708	**1.279**	0.112	-1.397

- **cWN L/S** outperform benchmarks in crash and bear markets, with **smaller drawdowns and higher sharpes.**
- **cWN L/S is the most robust**, achieving **positive and the highest Sharpe** in Bear 2022.
- Results suggest the models are effective at **downside risk and drawdown protection**.

Adding Transaction Cost



- At **10 bps cost with daily rebalancing**, gross signals are wiped out: all L/S strategies fall to about **-80%** cum. return.
- cWN gross Sharpe (0.726) collapses to **-2.44 net**, highlighting that daily turnover overwhelms the predictive signal.
- Key implication: **lower rebalancing frequency, signal thresholding, or turnover-penalizing optimization** are needed to make the strategy implementable.

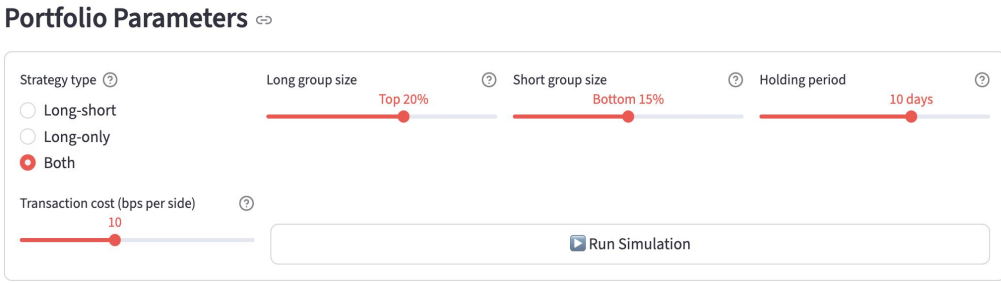
Streamlit Interactive Dashboard

https://ieor4733-final-project-ken5aqj7otchn4np6zrgwr.streamlit.app

Four Dashboard Pages

- 01 Data Explorer: feature distributions, ticker-level time series
- 02 Backtest Results: cumulative returns, rolling Sharpe, drawdown, annual returns
- 03 Regime Analysis: Bull / COVID Crash / Bear 2022 performance comparison
- 04 Run Simulation: interactive portfolio rules — long/short size, holding period, cost

No retraining required - Simulation page uses returns from previous model predictions

Simulation Example (biweekly rebalancing):

- At **10-day rebalancing**, net returns turn **positive** for the best model cWN and Sharpe rises vs. daily rebalancing
- Reducing turnover frequency is the single most impactful lever for making the strategy implementable